Xu Zihan

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Bachelor of Science (Honours) Computer Science student with hands-on experience in full-stack web development, data analysis, and machine learning. Skilled in modern web frameworks (Next.js, React, Django), REST APIs, and database management, with additional exposure to AI-powered applications and game development. Brings strong teamwork, communication, and event coordination experience from National Service and part-time roles, with a keen interest in pursuing a career in web development.

EDUCATION

Bachelor of Science (First Class Honours), Major in Computer Science

The University of London

Apr 2023 - Apr 2026

GCE 'A' Levels

Yishun-Innova Junior College

Feb 2019 - Nov 2020

GCE 'O' Levels

Manjusri Secondary School

Feb 2014 - Nov 2018

WORK EXPERIENCE

NVPC Software Engineer Intern (Power Automate, Python) **May 2026 - Jul 2026**

- Designed, built, and maintained multiple interconnected Power Automate flows across production systems for the Digital Innovation team, automating manual processes and conducting UAT testing for each automation operation before deployment.
- Built an email automation flow that sends personalised client emails driven by dynamic Excel sheet data.
- Developed a FormSG response automation flow that captures form submissions via email notifications and updates corresponding records in Excel, keeping downstream data in sync.
- Built a Python-based charity verification pipeline, cross-referencing data.gov.sg and NVPC's giving .sg datasets by using SQL and DuckDB to validate listed companies against official registries.
- Collaborated with other departments to understand their daily operations and identify automation opportunities, translating manual workflows into Power Automate solutions.
- Authored User Guide documentation for Power Automate flows to support adoption, troubleshooting, and handoff across departments.

Singapore Army, Administration Support Assistant (NS) **Apr 2021 - Apr 2023**

- Used Power BI to help with Presentation slides for Chief Safety Officer.
- Managed logistics, schedules, and resources to enhance event efficiency and outcomes.
- Coordinated team efforts to ensure seamless execution of events, resulting in high levels of satisfaction
- Implemented security measures during large events to prioritize attendee safety and well-being
- Assisted with ushering tasks, offering guidance to guests and maintaining order and compliance
- Demonstrated quick response to incidents, effectively resolving issues and maintain event security
- Contributed to the safeguarding and management of important documents, ensuring adherence to confidentiality and security standards

F&N Foods Singapore, Sales Associate **Feb 2021 - Apr 2021**

- Provided attentive customer service, ensuring satisfaction by addressing customer needs promptly

- Maintained product flow, filling, and presentation standards throughout the store

- Conducted product demonstrations and tastings to showcase features, benefits and unique selling points

- Monitored market trends, customer preferences and competitor activities to refine sales strategies accordingly.

SKILLS

Technical Skills:

- **Frontend**: Next.js, EJS, React, HTML/CSS, Tailwind CSS, JavaScript, Streamlit, Typescript
- **Backend**: Node.js, Express.js, Django, Django REST Framework, Flask, FastAPI
- **Databases**: SQLite, PostgreSQL, SQL (DuckDB), MongoDB
- **DevOps & Tools:** Git , Redis, Celery
- **AI & Machine Learning**: OpenAI API (RAG), BERT (NLP Transformers), PyTorch, scikit-learn (KNN)
- **Game Development:** Unity (URP), C#, WebGL Builds, 2D Sprite Rendering
- **APIs & Integration:** REST APIs, WebSockets (Django Channels)
- **Data Processing:** Pandas, NumPy, ETL Pipelines, CSV Processing
- **Authentication**: bcrypt, JWT, Session Management, Role-Based Access Control
- **Testing**: Django TestCase, Unit Testing, pytest
- **Office Tools:** Microsoft Office Suite , Power Automate, Excel, Powerpoint
- **CI/CD Tools**: Vercel, Railway, Docker

Soft Skills:

- Demonstrated proficiency in communication and interpersonal skills
- Strong customer service and guest relations capabilities
- Effective team leadership and collaboration skills
- Proven adaptability and flexibility in various contexts
- Commitment to customer satisfaction and needs-focused approach
- Experience in event coordination and logistics management

Projects

Music Recommendation App with Machine Learning | Python, Typescript, Docker, RAG (Llama 3.2 model), KNN, Pandas, MongoDB, FastAPI, Next.js, Tailwind CSS

- Built a content-based recommendation engine using scikit-learn K-Nearest Neighbors (KNN) with cosine similarity and StandardScaler using 11-dimensional normalized audio feature vectors retrieved via Spotify and Reccobeats APIs.
- Integrated a local LLM (Llama 3.2 1B Instruct) containerized with llama-cpp-python and Docker, implementing few-shot prompt engineering to translate mathematical user taste profiles into natural-language listening personality summaries.
- Built a full-stack REST application with a Next.js frontend and FastAPI backend, featuring JWT bearer authorization, MongoDB state persistence, dynamic track searching, and Genius API lyrics integration.
- Deployed backend on Railway and frontend on Vercel, with MongoDB Atlas for cloud database hosting

Full-Stack Social Platform with AI-Powered Chat | Next.js, React, SQLite, OpenAI API, bcrypt, Tailwind CSS, REST APIs

- Worked in a team of 3 developers to develop and deploy a sophisticated social networking application that uses Next.js as the main frontend framework, Git for version control and DevOps practices to ensure reliability, scalability and automated deployments.
- Implemented a RAG system using OpenAI API to power intelligent responses in the project.
- Developed React components including PostList (with like, save, share, comment functionality) and dynamic navbar with user-specific rendering
- Implemented dynamic chatroom system with message persistence and room-specific routing using Next.js dynamic parameters
- Created profile management (bio editing), post creation area, bookmark saving, and interaction history tracking
- Proficient in feature branching, git pull, git merge, git rebase, resolving merge conflicts, conducting code reviews via pull requests, and maintaining a clean commit history in team environments

E-Learning Platform | Django, Django REST Framework, Redis, Django Channels (WebSockets), Celery, SQLite/PostgreSQL, Django Views (CBVs), HTML/CSS/JS

- Built a full-featured Learning Management System (LMS) enabling course creation, enrollment, and interactive feedback systems with teacher/student role-based access control.
- Implemented a real-time course-specific chat rooms using redis-server and Django Channels.

Data Analysis Project | SQL, PostgreSQL/SQLite, CSV data processing, Database Design, ETL Pipelines

- Analyzed crime rate trends across the New South Wales Australian suburbs during the COVID-19 pandemic (2020-2022) by building a structured database from raw CSV crime data.
- Built a complete ETL workflow using ingest-data.sql and load-data.sql to extract data from raw CSV files.
- Processed and analyzed 90000 records from Australian Bureau of Statistics crime datasets.

Event Ticketing System with SQL Database | Node.js, Express.js, SQLite, REST APIs, EJS (Embedded JavaScript Templates), Bootstrap 5, HTML/CSS

- Built a full-stack event ticketing web application using Node.js and Express.js with modular routing architecture for organizers and attendees
- Designed and implemented RESTful API endpoints for event creation, booking management, attendee registration, and site settings using Express route handlers
- Designed relational SQLite database with optimized schema and parameterized queries for security
- Implemented dynamic server-side rendering using EJS templates for responsive UI across attendee and organizer dashboards

2D RPG Turn-Based Card Game | Unity (URP - Universal Render Pipeline), C#, WebGL, 2D Sprite-based Graphics, Turn-Based Game Mechanics

- Built and deployed 2D RPG card game where players collect cards, build decks, and engage in turn-based combat against AI opponents with strategic card-driven mechanics.
- Successfully built and deployed game to WebGL platform, enabling browser-based play without installation

Genetic Algorithm for 3D Locomotion| Python, PyBullet, NumPy, Genetic Algorithms, URDF Generation, Multiprocessing, Procedural Terrain

- Designed and implemented a genetic algorithm that evolves modular 3D walking creatures in a PyBullet physics simulator, each creature's morphology (link lengths, radii, masses, joint axes, origins) and motor control (waveform, amplitude, frequency) are encoded in a variable-length genome.
- Evolved creatures to traverse complex 3D terrains: flat ground, Gaussian pyramid, and a hard rocky mountain with obstacles (loaded from custom URDFs)

Power Automate Automation Systems | Power Automate, SharePoint, Excel, Email/HTML Templating

- Designed, built, and maintained multiple interconnected Power Automate flows across various production system.
- Debugged logic errors and resolved SharePoint file reference issues in multi-step flows, improving reliability of automated data pipelines end-to-end.
- Diagnosed and resolved email rendering inconsistencies across multiple email clients, standardizing HTML/CSS output for RSVP notification templates.
- Designed an email flow, extending the automation system's notification logic and scheduling capabilities.